

Footprinting

21 Sections Medium Offensive

This module covers techniques for footprinting the most commonly used services in almost all enterprise and business IT infrastructures. Footprinting is an essential phase of any penetration test or security audit to identify and prevent information disclosure. Using this process, we examine the individual services and attempt to obtain as much information from them as possible.

100% Completed



Attacking Common Applications

33 Sections Medium Offensive

Penetration Testers can come across various applications, such as Content Management Systems, custom web applications, internal portals used by developers and sysadmins, and more. It's common to find the same applications across many different environments. While an application may not be vulnerable in one environment, it may be misconfigured or unpatched in the next. It is important as an assessor to have a firm grasp of enumerating and attacking the common applications discussed in this module. This knowledge will help when encountering other types of applications during assessments.

100% Completed



Shells & Payloads

17 Sections Medium Offensive

Gain the knowledge and skills to identify and use shells & payloads to establish a foothold on vulnerable Windows & Linux systems. This module utilizes a fictitious scenario where the learner will place themselves in the perspective of a sysadmin trying out for a position on CAT5 Security's network penetration testing team.

100% Completed



Attacking Common Services

19 Sections Medium Offensive

Organizations regularly use a standard set of services for different purposes. It is vital to conduct penetration testing activities on each service internally and externally to ensure that they are not introducing security threats. This module will cover how to enumerate each service and test it against known vulnerabilities and exploits with a standard set of tools.

100% Completed



Web Attacks

18 Sections Medium Offensive

This module covers three common web vulnerabilities, HTTP Verb Tampering, IDOR, and XXE, each of which can have a significant impact on a company's systems. We will cover how to identify, exploit, and prevent each of them through various methods.

100% Completed



Information Gathering - Web Edition

19 Sections Easy Offensive

This module equips learners with essential web reconnaissance skills, crucial for ethical hacking and penetration testing. It explores both active and passive techniques, including DNS enumeration, web crawling, analysis of web archives and HTTP headers, and fingerprinting web technologies.

100% Completed



File Upload Attacks

11 Sections Medium Offensive

Arbitrary file uploads are among the most critical web vulnerabilities. These flaws enable attackers to upload malicious files, execute arbitrary commands on the back-end server, and even take control over the entire server and all web applications hosted on it and potentially gain access to sensitive data or cause a service disruption.

100% Completed



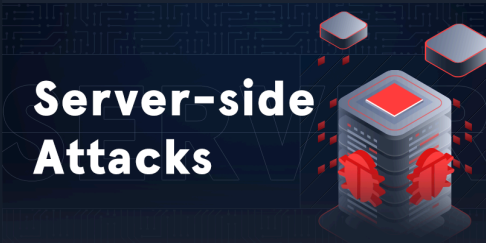


Active Directory Enumeration & Attacks

36 Sections Medium Offensive

Active Directory (AD) is the leading enterprise domain management suite, providing identity and access management, centralized domain administration, authentication, and much more. Due to the many features and complexity of AD, it presents a large attack surface that is difficult to secure properly. To be successful as infosec professionals, we must understand AD architectures and how to secure our enterprise environments. As Penetration testers, having a firm grasp of what tools, techniques, and procedures are available to us for enumerating and attacking AD environments and commonly seen AD misconfigurations is a must.

100% Completed



Server-side Attacks

19 Sections Medium Offensive

A backend that handles user-supplied input insecurely can lead to devastating security vulnerabilities such as sensitive information disclosure and remote code execution. This module covers how to identify and exploit server-side bugs, including Server-Side Request Forgery (SSRF), Server-Side Template Injection (SSTI), and Server-Side Includes (SSI) injection attacks.

100% Completed



Password Attacks

26 Sections Medium Offensive

Passwords are still the primary method of authentication in corporate networks. If strong password policies are not enforced, users often choose weak, easy-to-remember passwords that can be cracked offline and leveraged to escalate access. As penetration testers, we encounter passwords in many forms during our assessments. It's essential to understand how passwords are stored, how they can be retrieved, methods for cracking weak passwords, techniques for using hashes that cannot be cracked, and how to identify weak or default password usage.

84.62% Completed



Session Security

14 Sections Medium Offensive

Maintaining and keeping track of a user's session is an integral part of web applications. It is an area that requires extensive testing to ensure it is set up robustly and securely. This module covers the most common attacks and vulnerabilities that can affect web application sessions, such as Session Hijacking, Session Fixation, Cross-Site Request Forgery, Cross-Site Scripting, and Open Redirects.

100% Completed



Pivoting, Tunneling, and Port Forwarding

18 Sections Medium Offensive

Once a foothold is gained during an assessment, it may be in scope to move laterally and vertically within a target network. Using one compromised machine to access another is called pivoting and allows us to access networks and resources that are not directly accessible to us through the compromised host. Port forwarding accepts the traffic on a given IP address and port and redirects it to a different IP address and port combination. Tunneling is a technique that allows us to encapsulate traffic within another protocol so that it looks like a benign traffic stream.

100% Completed



Web Service & API Attacks

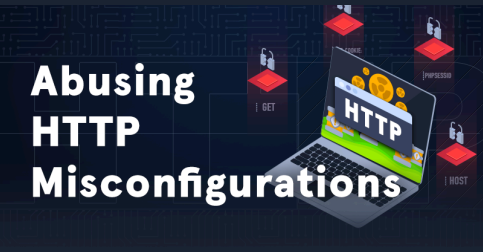
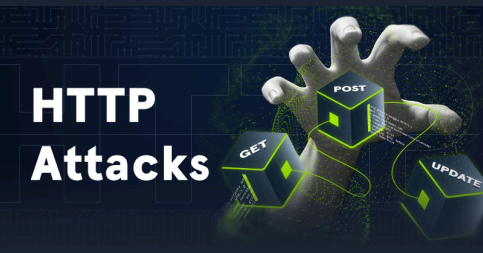
13 Sections Medium Offensive

Web services and APIs are frequently exposed to provide certain functionalities in a programmatic way between heterogeneous devices and software components. Both web services and APIs can assist in integrating different applications or facilitate separation within a given application. This module covers how to identify the functionality a web service or API offers and exploit any security-related inefficiencies.

100% Completed



 Bug Bounty Hunting Process	Bug Bounty Hunting Process 6 Sections Easy General Bug bounty programs encourage security researchers to identify bugs and submit vulnerability reports. Getting into the world of bug bounty hunting without any prior experience can be a daunting task, though. This module covers the bug bounty hunting process to help you start bug bounty hunting in an organized and well-structured way. It's all about effectiveness and professionally communicating your findings.	100% Completed
 Documentation and Reporting	Documentation & Reporting 8 Sections Easy General Proper documentation is paramount during any engagement. The end goal of a technical assessment is the report deliverable which will often be presented to a broad audience within the target organization. We must take detailed notes and be very organized in our documentation, which will help us in the event of an incident during the assessment. This will also help ensure that our reports contain enough detail to illustrate the impact of our findings properly.	100% Completed
 Attacking Enterprise Networks	Attacking Enterprise Networks 14 Sections Medium Offensive We often encounter large and complex networks during our assessments. We must be comfortable approaching an internal or external network, regardless of the size, and be able to work through each phase of the penetration testing process to reach our goal. This module will guide students through a simulated penetration testing engagement, from start to finish, with an emphasis on hands-on testing steps that are directly applicable to real-world engagements.	100% Completed
 Introduction to Deserialization Attacks	Introduction to Deserialization Attacks 15 Sections Hard Offensive In this module, we will explore deserialization attacks with specific examples in Python and PHP.	100% Completed
 Attacking Authentication Mechanisms	Attacking Authentication Mechanisms 20 Sections Medium Offensive Authentication plays an essential role in almost every web application. If a vulnerability arises in the application's authentication mechanism, it could result in unauthorized access, data loss, or potentially even remote code execution, depending on the application's functionality. This module will provide an overview of various access control methods, such as JWT, OAuth, and SAML, and potential attacks against each.	100% Completed
 Introduction To NoSQL Injection	Introduction to NoSQL Injection 12 Sections Medium Offensive In this module, we will look at exploiting NoSQL injection vulnerabilities, specifically MongoDB, with examples in Python, PHP, and Node.JS.	100% Completed
 Blind SQL Injection	Blind SQL Injection 16 Sections Hard Offensive In this module, we cover blind SQL injection attacks and MSSQL-specific attacks.	100% Completed
 HTTPs/TLS Attacks	HTTPs/TLS Attacks 15 Sections Medium Offensive This module covers details on Transport Layer Security (TLS) and how it helps to make HTTP secure with the widely used HTTPS. That includes how TLS works, how TLS sessions are established, common TLS misconfigurations, as well as famous attacks on TLS. We will discuss how to identify, exploit, and prevent TLS attacks.	100% Completed

	Advanced SQL Injections 12 Sections Hard Offensive This module covers advanced SQL injection techniques with a focus on white-box testing, Java/Spring and PostgreSQL.	100% Completed
	Abusing HTTP Misconfigurations 20 Sections Hard Offensive This module covers three common HTTP vulnerabilities: Web Cache Poisoning, Host Header Vulnerabilities, and Session Puzzling or Session Variable Overloading. These vulnerabilities can arise on the HTTP level due to web server misconfigurations, other systems that have to be considered during real-world deployment such as web caches, or coding mistakes in the web application. We will cover how to identify, exploit, and prevent each of these vulnerabilities.	100% Completed
	HTTP Attacks 18 Sections Hard Offensive This module covers three HTTP vulnerabilities: CRLF Injection, HTTP Request Smuggling, and HTTP/2 Downgrading. These vulnerabilities can arise on the HTTP level in real-world deployment settings utilizing intermediary systems such as reverse proxies in front of the web server. We will cover how to identify, exploit, and prevent each of these vulnerabilities.	100% Completed
	Whitebox Attacks 15 Sections Hard Offensive This module explores several web vulnerabilities from a whitebox approach: Prototype Pollution, Timing Attacks & Race Conditions, and those arising from Type Juggling. We will discuss how to identify, exploit, and prevent each vulnerability.	100% Completed
	Injection Attacks 15 Sections Medium Offensive This module covers three injection attacks: XPath injection, LDAP injection, and HTML injection in PDF generation libraries. While XPath and LDAP injection vulnerabilities can lead to authentication bypasses and data exfiltration, HTML injection in PDF generation libraries can lead to Server-Side Request Forgery (SSRF), Local File Inclusion (LFI), and other common web vulnerabilities. We will cover how to identify, exploit, and prevent each of these injection attacks.	100% Completed
	Introduction to Threat Hunting & Hunting With Elastic 6 Sections Medium Defensive This module initially lays the groundwork for understanding Threat Hunting, ranging from its basic definition, to the structure of a threat hunting team. The module also dives into the threat hunting process, highlighting the interrelationships between threat hunting, risk assessment, and incident handling. Furthermore, the module elucidates the fundamentals of Cyber Threat Intelligence (CTI). It expands on the different types of threat intelligence and offers guidance on effectively interpreting a threat intelligence report. Finally, the module puts theory into practice, showcasing how to conduct threat hunting using the Elastic stack. This practical segment uses real-world logs to provide learners with hands-on experience.	16.67% Completed
	Modern Web Exploitation Techniques 18 Sections Hard Offensive This module covers advanced web concepts and exploitation techniques, including performing DNS Rebinding to bypass faulty SSRF filters and the Same-Origin Policy, identifying and exploiting Second-Order vulnerabilities, and conducting common web attacks via WebSocket connections.	100% Completed

Advanced XSS and CSRF Exploitation



Advanced XSS and CSRF Exploitation

17 Sections

Medium

Offensive

Modern web browsers and applications utilize a variety of security measures to protect against CSRF and XSS vulnerabilities, rendering their exploitation more difficult. This module focuses on exploiting advanced CSRF and XSS vulnerabilities, identifying and bypassing weak and wrongly implemented defensive mechanisms.

100% Completed

Parameter Logic Bugs



Parameter Logic Bugs

21 Sections

Hard

Defensive

This 'secure coding' module teaches how to identify logic bugs through code review and analysis, and covers three types of logic bugs caused by user input manipulation.

100% Completed

Advanced Deserialization Attacks



Advanced Deserialization Attacks

13 Sections

Hard

Offensive

This module focuses on developing custom exploits for .NET deserialization vulnerabilities from a whitebox perspective.

100% Completed

Intro to Whitebox Pentesting



Intro to Whitebox Pentesting

18 Sections

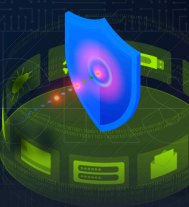
Hard

Offensive

Whitebox penetration testing enables thorough testing to identify various hard-to-find vulnerabilities. This module covers the process of whitebox pentesting and follows that with a practical demo by exploiting an advanced code injection vulnerability.

100% Completed

API Attacks



API Attacks

13 Sections

Medium

Offensive

Web APIs serve as crucial connectors across diverse entities in the modern digital landscape. However, their extensive functionality also exposes them to a range of potential attacks. This module introduces API Attacks, with a specific focus on the OWASP API Security Top 10 - 2023.

100% Completed

Attacking GraphQL



Attacking GraphQL

9 Sections

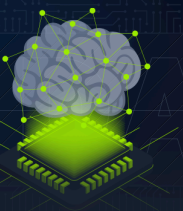
Medium

Offensive

GraphQL is a query language for APIs as an alternative to REST APIs. Clients are able to request data through GraphQL queries. If improperly configured or implemented, common web security vulnerabilities such as Information Disclosure, SQL Injection, and Insecure Direct Object Reference (IDOR) may arise.

100% Completed

Fundamentals of AI



Fundamentals of AI

24 Sections

Medium

General

This module provides a comprehensive guide to the theoretical foundations of Artificial Intelligence (AI). It covers various learning paradigms, including supervised, unsupervised, and reinforcement learning, providing a solid understanding of key algorithms and concepts.

16.67% Completed